TenderBlock - Frontend Design Document
Frontend handoff | TenderBlock | 7 November 2026

1. DESIGN GOALS

Create a role-aware, accessible frontend that follows actual API states and uses a typed API boundary. Backend authorization remains authoritative; UI route guards are convenience only.

2. APPLICATION STRUCTURE

Use shared public/auth pages and authenticated shell, then role route groups /admin, /authority, /contractor, /evaluator, /auditor. Suggested layers: pages, shared components, API client, DTO types, session store, validation utilities. Follow the repository's existing framework and conventions.

3. ROUTE AND NAVIGATION MODEL

Public: login and registration. Authenticated shell selects role dashboard from auth response. Show accountStatus prominently. Pending onboarding should permit profile creation/update for Authority/Contractor. Do not offer ADMIN registration. Logout clears session.

4. PAGE INVENTORY

Login/register: role selection, validation, pending approval state. Shared profile for Authority/Contractor. Admin: pending/all users, user details, account actions, tender review and evaluator selection. Authority: tender list/detail/create/edit/cancel. Contractor: published tender browser/detail, bid wizard, upload, submit, own bids/detail/download. Evaluator: assignments, open bids, bids and PDFs, evaluation editor/finalize, award summary. Auditor: report queue/detail/claim/resolve and tender view.

5. STATUS-AWARE ACTIONS

Tender: SUBMITTED, PUBLISHED, REJECTED, UNDER_EVALUATION, AWARDED, CANCELLED, VOIDED. Bid: DRAFT, SUBMITTED, WITHDRAWN, OPENED, UNDER_EVALUATION, EVALUATED, SELECTED, NOT_SELECTED. Evaluation: DRAFT, FINALIZED. Report: OPEN, UNDER_REVIEW, RESOLVED, DISMISSED. Hide invalid actions but still handle API conflicts. No withdrawal action exists despite enum value.

6. API AND SESSION DESIGN

Base path /api/v1. Public login/register; other app APIs require Bearer token. JWT expires after one hour; no refresh endpoint. Clear session and route to login on 401. Explain 403 as role/account/state issue; 409 as state conflict. Avoid depending on error body schema. Use page,size,sort for Spring Page routes; display total pages/items. No maximum page size exists server-side.

7. DATA COMPONENT PATTERNS

UUIDs are strings. BigDecimal values should be handled as decimal strings, not binary floating point. Timestamps are ISO-8601 Instants. Paginated response is Spring Page with content, pageable, totalElements, totalPages, size, number, sort, etc. Upload FormData field must be named file; let browser set multipart boundary. Download PDF as bytes using Content-Disposition filename.

8. UX AND ACCESSIBILITY

Semantic headings and labels, keyboard-accessible dialogs, visible focus and inline validation. Confirm tender cancellation, bid submission, evaluation finalization, report resolution. Show deadlines in local time with timezone. Provide loading, empty, retry, and validation states. Do not expose JWTs or raw internal exceptions.

9. SECURITY

Do not integrate /api/blockchain/** into dashboards; currently public test read/write endpoints. Do not treat UI role/user ID as identity. Never log token or proposal content. Use secure token storage for chosen platform and clear on logout. Coordinate deployment CORS

origins with backend.

10. INTEGRATION CAVEATS

Admin/auditor tender detail currently has broken role matching. Cancellation can return 500 for expected failures. Account-state enforcement is inconsistent for stale tokens. No CORS configuration found. No dedicated verified-evaluator listing/search endpoint exists. Report generation service is not wired to a caller.

11. TEAM DECISIONS

Confirm frontend stack/design system, deployment origins, token storage, evaluator selection UX, whether non-company roles need profile screens, public tender detail/award visibility, and backend fixes needed before release.